

CPU vs GPU & Microprocessor

Lesson 06 — Serial vs Parallel, Microprocessor vs Microcontroller

JKSSB Computer Awareness

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Video Lesson 06

Watch: CPU vs GPU: What's the Difference?

Serial vs Parallel Processing

- ▶ **Serial processing:** one task is carried out at a time, in sequence.
- ▶ **Parallel processing:** several tasks or operations proceed at the same time.
- ▶ A few fast cores suit serial work; many cores working together suit parallel work.
- ▶ The office desktop (EX-01) runs several programs together, so it depends on parallel support.

Exam cue

Serial = order (A, then B, then C). Parallel = together (A and B and C).

CPU: Few Powerful Cores

- ▶ The CPU is the **Central Processing Unit**, the general-purpose processor.
- ▶ It has **few, powerful cores**, each able to run a sequential instruction stream quickly.
- ▶ Design goal: **low latency** — finish one task with minimum delay.
- ▶ In EX-01, the CPU runs the operating system, MS Word, and spreadsheet logic step by step.

Remember

A CPU core is general-purpose: it can run any instruction, not only graphics.

GPU: Many Simple Cores

- ▶ GPU is the **Graphics Processing Unit**.
- ▶ It has **many simpler cores**, each performing a small amount of work.
- ▶ Design goal: **high throughput** — a huge number of parallel operations per second.
- ▶ Natural fit: pixels and graphics, plus scientific and machine-learning workloads.

Trap

GPU cores are not general-purpose CPU cores. They are many, simpler, and optimised for parallel work.

CPU vs GPU at a Glance

Feature	CPU vs GPU	
Cores	CPU: few, powerful	GPU: many, simple
Optimised for	CPU: latency	GPU: throughput
Best at	CPU: sequential	GPU: parallel
Example	CPU: desktop logic	GPU: pixels

Think: a few experts (CPU) vs a large team of simple workers (GPU).

Cores vs Threads and Hyper-Threading

Core

- ▶ A physical processing unit.
- ▶ A quad-core CPU has four such units.

Thread

- ▶ A stream of instructions a core executes.
- ▶ **Hyper-threading**: one core presents two logical threads.

Do not confuse

Two logical threads on one core are **not** two physical cores.

Cache Depth: L1, L2, L3

- ▶ Cache is small, fast memory held close to the core.
- ▶ **L1**: smallest and fastest, nearest the core.
- ▶ **L2**: larger and a little slower than L1.
- ▶ **L3**: largest and slowest of the three, shared across cores.
- ▶ Speed order: L1 faster than L2 faster than L3, then main memory.

Exam cue

Smallest → fastest → closest: **L1 < L2 < L3**.

Clock Speed and Multi-Core

- ▶ **Clock speed** is measured in **GHz** (gigahertz): billions of cycles per second.
- ▶ A higher clock speed generally means more work completed per second, all else equal.
- ▶ A **multi-core** processor has several cores on one chip, which enables parallel work.
- ▶ Clock speed alone does not decide real speed — cores, cache, memory, and I/O all matter.

Microprocessor: A CPU on a Chip

- ▶ A **microprocessor** is a **central processing unit implemented on a single chip**.
- ▶ It is a **general-purpose** processor: software decides what it does.
- ▶ It needs external memory and I/O devices to form a complete working system.
- ▶ The office desktop (EX-01) is built around a microprocessor.

Definition

Microprocessor = the CPU, integrated on one chip.

Microcontroller: Integration on One Chip

- ▶ A **microcontroller** adds more onto one chip: a microprocessor **plus** memory **plus** I/O peripherals.
- ▶ It forms a near-complete computer on one chip, used for **embedded/special-purpose** control.
- ▶ Everyday uses: washing machines, remote controls, and appliance or vehicle controllers.
- ▶ The key difference from a microprocessor is the **level of integration**, not clock speed.

Microprocessor vs Microcontroller

Microprocessor	Microcontroller
CPU on a chip	CPU + memory + I/O on a chip
General-purpose	Embedded / special-purpose
Needs external parts	Self-contained system
E.g. office desktop	E.g. appliance controller

Trap

Do not call a microcontroller “just a microprocessor” — it is a more integrated device.

Diagnosing a “Slow PC” (EX-01)

- ▶ EX-01: a quad-core 2.4 GHz desktop with 16 GB RAM and an SSD.
- ▶ Yet one program still runs slowly.
- ▶ More cores do not help a **single-threaded** program.
- ▶ The cause is usually a **bottleneck**: memory, disk I/O, or software that is not multithreaded.

Trap (TRAP-11)

A slow program is not automatically a “slow CPU.” Diagnose the stated bottleneck first.

Lesson 06: Recall

- ▶ CPU: few powerful, general-purpose cores optimised for latency.
- ▶ GPU: many simple cores optimised for parallel throughput.
- ▶ Serial = one at a time; parallel = many at once.
- ▶ Microprocessor = CPU on a chip; microcontroller = CPU + memory + I/O on one chip.
- ▶ Quad-core but slow → suspect a bottleneck, not always the CPU.